Interference

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Argument

- In **Discrete Mathematics**, an **argument** is a sequence of **statements (propositions)** in which:
  - Some statements are given as **premises** (assumptions or evidence).
  - One statement is the **conclusion**, which is claimed to follow logically from the premises.

Argument = Premises + Conclusion

- An argument is said to be **valid** if the conclusion logically follows from the premises (i.e., if the premises are true, then the conclusion must also be true).

Example

- **Example 1:**
  1. **Premise 1:** If a number is even, then its square is even.
  2. **Premise 2:** 4 is an even number.
  3. **Conclusion:** Therefore, 4^2 (=16) is even.

- **Example 2 (classic logic form):**
  1. **Premise 1:** All humans are mortal.
  2. **Premise 2:** Socrates is a human.
  3. **Conclusion:** Therefore, Socrates is mortal.

Form of Logical Arguments

$$\begin{array}{c} Premise\ 1 \\ Premise\ 2 \\ \vdots \\ Premise\ \mathrm{n} \\ \hline \therefore Conclusion \end{array}$$

OR

$$Premise\ 1, Premise\ 2, \ldots, Premise\ 2 \mapsto Conclusion$$

$$P_1, P_2, \ldots, P_n \Rightarrow C$$

$$\begin{array}{l} \text{If Marcus is a poet, then he is poor.} \\ \text{Marcus is a poet.} \\ \hline \therefore\ \text{Marcus is poor.} \end{array}$$

Rules of Interference

- In **Discrete Mathematics (Logic)**, the **rules of inference** are the formal rules that describe how you can derive a valid conclusion from given premises.

- Rules of Inference provide the **templates or guidelines** for constructing valid arguments from the statements that we already have.

Rule	Tautology	Name
$p \to q$ p $\therefore q$	$((p \to q) \wedge p) \Rightarrow q$	Modus Ponens (Law of Detachment)
$p \to q$ $\neg q$ $\therefore \neg p$	$((p \to q) \wedge \neg q) \Rightarrow \neg q$	Modus Tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \Rightarrow (p \to r)$	Hypothetical Syllogism (Transitivity)
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \Rightarrow q$	Disjunctive Syllogism
p $\therefore p \vee q$	$p \Rightarrow p \vee q$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \Rightarrow p$	Simplification
p q $\therefore p \wedge q$	$(p) \wedge (q) \Rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \Rightarrow (q \vee r)$	Resolution

Modus Popens

If a conditional statement ("if-then" statement) is true, and its antecedent (the "if" part) is true, then its consequent (the "then" part) must also be true.

Form: If p $\longrightarrow$ q and p, then q.

"If it is raining, then I will make tea."	$p \rightarrow q$	**Modus Ponens (Law of Detachment)**
"It is raining."	p	$\left(\left(p \rightarrow q\right) \wedge p\right) \Rightarrow q$
"Therefore, I will make tea."	$\therefore q$	

Modus Tollens (Law of Contrapositive)

If a conditional statement is true, and its consequent is false, then its antecedent must also be false.

Form: If p $\longrightarrow$ q and ¬q, then ¬p.

"If it is raining, then I will make tea."	$p \rightarrow q$	**Modus Tollens**
"I do not make tea."	$\neg q$	$\left(\left(p \rightarrow q\right) \wedge \neg q\right) \Rightarrow \neg q$
"Therefore, it is not raining."	$\therefore \neg p$	

Hypothetical Syllogism

If two conditional statements are true, where the consequent of the first is the antecedent of the second, then a third conditional statement combining the antecedent of the first and the consequent of the second is also true.

Form: If p → q and q → r, then p → r.

"If it is raining, then I will make tea." $p \rightarrow q$

"If i make tea, then I will read a book." $q \rightarrow r$

"Therefore, if it rains, then I will read a book." $\therefore p \rightarrow r$

Hypothetical Syllogism

$$\left(\left(p \rightarrow q\right) \wedge \left(q \rightarrow r\right)\right) \Rightarrow \left(p \rightarrow r\right)$$

Disjunctive Syllogism

If a disjunction (an "or" statement) is true, and one of the disjuncts (the parts of the "or" statement) is false, then the other disjunct must be true.

Form: If q $\lor$ r and ¬q, then r.

"I will make tea, or I will read a book."	$q \lor r$	**Disjunctive Syllogism**
"I will not make tea."	$\neg q$	$\big((q \lor r) \land \neg q\big) \Rightarrow r$
"Therefore, I will read a book."	$\therefore r$	

Addition

If a statement is true, then the disjunction (an "or" statement) of that statement with any other statement is also true.

Form: If q, then q $\vee$ r

"I will make tea."	q	**Addition**
"Therefore, I will make tea, or I will read a book."	$\therefore q \vee r$	$q \Rightarrow q \vee r$

Simplification

If a conjunction (an "and" statement) is true, then each of its conjuncts is also true.

Form: If q ∧ r, then q

It is raining and windy.
It is raining.

$$q \wedge r$$
$$\therefore q$$

Simplification

$$\left(p \wedge q\right) \Rightarrow p$$

Conjunction

If two statements are true, then their conjunction (an "and" statement) is also true.
Form: If p and q, then p ∧ q.

- Premise: It is raining.
- Premise: It is windy.
- Conclusion: It is raining and windy.

$$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$$

$$(p) \wedge (q) \Rightarrow (p \wedge q)$$

Conjunction

Resolution

If two disjunctions ("or" statements) are true, and one contains a proposition (P) while the other contains its negation (¬P), then the disjunction of the remaining parts is true.

Form: *If P∨Q and ¬P∨R, then Q∨R.*

$$\begin{array}{c} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array} \qquad \Big| \qquad ((p \vee q) \wedge (\neg p \vee r)) \Rightarrow (q \vee r) \qquad \Big| \qquad \text{Resolution}$$

Example:

- *Premise 1: It is raining* ***or*** *it is snowing.*
- *Premise 2: It is* ***not*** *raining* ***or*** *it is cold.*
- *Conclusion: It is snowing* ***or*** *it is cold.*